

George Webber

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Third-year PhD candidate at King's College London, specializing in inverse problems and generative machine learning and their application to medical imaging. Strong record of publication in top-tier journals and conferences. Holds a First-Class Integrated Master's degree in Mathematics & Computer Science from the University of Oxford. Research aims to advance medical image quality through innovative integration of machine learning methods.

Education

King's College London

Sept 2023 – Present

PhD in Biomedical Engineering & Imaging Sciences

- Project: *Fully Bayesian 3D PET-MR Neuroimaging Reconstruction*.
- Investigating pre-trained diffusion models as generative priors for inverse problems in medical imaging.
- Supervisors: Prof. Andrew J. Reader and Prof. Andrew P. King.
- Award: Won the faculty-level PGR Research Excellence Prize 2025.

St Anne's College, University of Oxford

Oct 2019 – Jun 2023

MMathCompSci – Mathematics & Computer Science

- First Class overall (77 %) and Distinction in final year (81 %).
- Master's thesis: *Improving the Security of Smartwatch Payment with Deep Learning*.

John Mason School, Abingdon

Sept 2012 – 2019

A-Levels and EPQ

- A*A*A*A*A* in Mathematics, Further Mathematics, Computer Science, Physics, and an EPQ.

Experience

Graduate Teaching Assistant

London, UK

King's College London

Nov 2023 – Present

- Deliver tutorials, labs and marking for undergraduate mathematics and medical imaging courses.

Alpha Tech Intern – Quant Analysis

London, UK

Man Group

Jun – Sep 2022

- Designed experiments on simulated portfolios, enhancing Value-at-Risk calculations by incorporating earnings-announcement effects.
- Co-authored an internal research paper and delivered a 30-minute presentation to senior risk managers.
- Shipped production-ready Python code to calculate daily risk metrics.

Senior Consultant

Oxford, UK

Oxford Strategy Group Digital

Sep – Dec 2021

- Open-sourced and extended Wise's variational-autoencoder model for customer-behaviour prediction.
- Authored tutorials and Jupyter notebooks demonstrating applications for fintech clients.

Software Engineering Intern

Remote / UK

Microsoft

Jul – Aug 2021

- Built and documented an automated monitoring solution for critical infrastructure.
- Acted as technical lead among interns—wrote high-level design docs, test plans, and conducted code reviews.

Selected Publications

Journal Articles

Personalized MR-Informed Diffusion Models for 3D PET Image Reconstruction

Aug 2025

G. Webber et al., *IEEE Transactions on Radiation and Plasma Medical Sciences*

doi:10.1109/TRPMS.2025.3602262 [↗](#)

- Proposes a novel personalization scheme for generating subject-specific training data, for the purpose of training a diffusion model that reconstructs PET images with superior accuracy.

Likelihood-Scheduled Score-Based Generative Modeling for Fully 3-D PET Image Reconstruction

June 2025

G. Webber et al., *IEEE Transactions on Medical Imaging*

doi:10.1109/TMI.2025.3576483 [↗](#)

- Proposes an algorithm for aligning the conditioning of a diffusion model with clinically-used algorithms, improving reconstruction speed and ease of hyperparameter optimization.

Diffusion Models for Medical Image Reconstruction

Sept 2024

G. Webber & A. J. Reader, *BJR—Artificial Intelligence*

doi:10.1093/bjrai/ubae013 [↗](#)

- Explains and surveys how diffusion models are used as a generative prior for medical image inverse problems.
- Most-read article in the journal (as of October 2025).

Conference Proceedings

Supervised Diffusion-Model-Based PET Image Reconstruction

Sept 2025

G. Webber et al., *proc. MICCAI 2025*, Daejeon, South Korea

arXiv:2506.24034 [↗](#)

- Demonstrates new methodology for conditioning a diffusion model directly on raw PET data, and shows this supervised methodology outperforms equivalent unsupervised approaches.

Submitted Manuscripts

Distributional Consistency Loss: Beyond Pointwise Data Terms in Inverse Problems

Submitted: September 2025

G. Webber & A. J. Reader, *ICLR 2026 (under review)*

- Formulated and validated a new data fidelity loss that overcomes the tendency of classical losses to overfit during long training, achieving more robust reconstructions and synergistic combination with regularization.

Technologies

Programming: Proficient with Python (NumPy, Pandas, SciPy; PyTorch, TensorFlow, scikit-learn; Matplotlib, Napari). Familiar with Java, C#, Scala, HTML, SQL, JavaScript, CSS, Haskell.

Tools & Workflow: Git / GitHub, Linux & Windows, Azure + GCS storage, Scrum / Agile (Confluence, Jira), LaTeX.

Activities

- Won sponsor prize at **Oxford Hack 2022** for novel LiDAR-data visualisation for Oxa autonomous vehicles.
- Elected **Student Representative** for the Smart Medical Imaging CDT cohort – organise research-development seminars and social events.
- Play tennis at a university level, representing King's at inter-university league matches.